Ankit Raj

PhD Candidate, Spatial Ecology & Paleontology lab
Quantitative Biosciences, Earth & Atmospheric Sciences
Georgia Institute of **Tech**nology, Atlanta, Georgia, USA

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Education

2024-Present

PhD in Quantitative Biosciences, Home school: Earth & Atmospheric Sciences

Georgia Institute of Technology, Atlanta, USA

Supervisor: **Dr. Jenny McGuire**, school of Biological Sciences, and school of Earth & Atmospheric Sciences

2018-2023

Five-Year BS-MS Dual Degree, **Major: Geological Sciences**

Indian Institute of Science Education & Research (IISER) Kolkata, India

CGPA: 8.57/10; **Major CGPA: 9.17/10**

Recipient of **the Chairman's Medal of Excellence for the best all-around performance**

Scientific Interests

- Wildfire regime & risk
- Remote sensing of Fire
- Landscape Connectivity Modeling
- Paleofire & Long term Ecological Change
- Uncertainty quantification
- Models of Ecological Diversification

Professional Service

Early Career Representative, International Paleofire Network (IPN), 2025–present.

Member, Methods Working Group of Paleofire Analysis.

Participant, PalaeOpen and **Fire Stakeholder** Workshop, Goslar, Germany, 12–16 March 2027.

Research Experience

2024- Present

PhD Dissertation

Fire and Mammal Communities through Time: Past, Present and Future

Supervisor: **Prof. Jenny McGuire**, Department of Biology and EAS, Gatech, Atlanta

- Quantifying how fire, climate, and human activity shaped mammal communities in the Greater Yellowstone Ecosystem over 17,000 years using multi-site charcoal and fossil records, and extending the analysis across North America
- Reconstructing fire history and mammal change from a single cave deposit (Natural Trap Cave, WY) using charcoal peak detection and Bayesian age–depth modeling
- Comparing past, present, and projected fire regimes using paleofire records and **satellite data** (MTBS, MODIS burned area and tree cover) to test whether mammal–vegetation relationships hold under future fire
- **Applied fire management:** modeling fire-spread and **wildlife-movement connectivity** to design **firebreak corridors** that limit large fires while keeping wildlife movement pathways open

First Year PhD Project

Disruption in the Trait-Environment relationship in African large mammals

Supervisor: **Prof. Jenny McGuire**, Department of Biology and EAS, Gatech, Atlanta

In Collaboration with **Dr. Michelle Lawing**, Department of Ecology and Conservation Biology, University of Texas A&M

- Quantified trait- environment relationship using ecometrics modelling
- Evaluated the point connectivity using Omniscape connectivity modelling
- Developed methods of analyzing the ecometric anomaly significance

MS Thesis Project

2022-2023

Patterns of Ecospace Utilization in Marine Invertebrates During the Mesozoic

Supervisor: **Dr. Subhronil Mondal**; Department of Earth Sciences, IISER Kolkata

In Collaboration with **Prof. Phillip Novack Gottshall**, Department of Biological Sciences, Benedictine University, Illinois

- Quantified ecological diversity using a theoretical ecospace framework.
- Compared patterns of ecospace occupation through time.
- Analyzed ecospace structuring in the aftermath of extinction events.
- Functional trait data collected from **Paleobiology database**

2018–2022

Earlier Research, IISER Kolkata

Mentors: **Dr. Subhronil Mondal, Dr. Gopala Krishna Darbha, Prof. Prasant Panigrahi**; IISER Kolkata

- **Models of ecological diversification (2022):** simulated diversification dynamics with the R package Ecospace; tested model selection using AIC, classification trees, and random forests.
- **Paleoecological diversification metrics (2021):** literature survey on ecological disparity and functional diversity metrics.
- **Microplastics detection (2019):** FTIR and NIR spectroscopy to detect ocean microplastics; ICP-MS for elemental identification.
- **Quantum computing (2018–19):** simulated quantum bomb detection on an IBM quantum computer.

Computational Skills

Programming Languages: R, Python,MATLAB; **Parallel computing**, Shell scripting: Bash, Quantum Computation (IBM), GIS, Origin, OS: Linux (Arch & Debian), **Supercomputing (HPC)**

Publications & Talks

"Demonstration of Quantum bomb detection using IBM Quantum computer" **A. Raj**, B. Das, B. K. Behera, and P. K. Panigrahi; Pre-print.

“Quantum bomb detection using IBM Quantum computer.” Quantum Information and Quantum Technology (QIQT) conference, IISER Kolkata, 2019.

Contributed Abstracts / Posters

“Disruptions to the trait-environment relationships of African mammals” **A. Raj**, D. A. Lauer, J. L. McGuire, and A. M. Lawing; Poster, The International Biogeography Society (TIBS) Conference, Aarhus, Denmark, 2026.

"Disruptions to the trait-environment relationships of African mammals" **A. Raj**, D. A. Lauer, J. L. McGuire, and A. M. Lawing; Poster, Gordon Research Seminar and Conference (GRC), Ventura, California, 2025.

Awards & Grants

EAS Travel Award; School of Earth & Atmospheric Sciences, Georgia Tech, $1,600, 2025.

CoS-VIGOR Travel Award: A Coaching Program to benefit graduate students and faculty; Georgia Tech, $3,000, 2024.

Chairman's Medal of Excellence for the best all-around performance, BS-MS batch of 2018–23; IISER Kolkata, 2023.

Conference Workshops & Additional Courses

2026	The International Biogeography Society Conference, Aarhus, Denmark (presented poster)
2026	International Paleofire Network Workshop, Penn State University, Pennsylvania
2025	Gordon Research Seminar and Conference, Ventura, California (presented poster)
2025	Paleontological Fieldwork at Natural Trap Cave (focus on mammalia), Wyoming
2025	Evolution Conference, University of Georgia, Athens, USA
2025	Paleobiology Database (PBDB) Workshop, George Mason University
2021–23	**Additional training & fieldwork:** Paleontology and stratigraphy fieldwork, Dwarka, Gujarat, India (2023); "Extinctions: Past, Present, & Future," Coursera, Emory University (2022); "Fundamentals of Paleontology," Kujju Field Training Center, Geological Survey of India Training Institute (2021).
2019	**Science Communication** Course; Prof. Lewis Dartnell (University of Westminster, London, UK), IISER Kolkata.

Teaching Experience

Fall 2026	GT 6000 Student Group Leader; Georgia Tech's small-group, peer-led, extended-orientation program for first-semester graduate students
Fall 2025	Graduate Teaching Assistant, Intro to Environmental Sciences (3 sections 24, 18, 24 students)
Spring 2025	Graduate Teaching Assistant, Earth Processes (19 students)
Fall 2024	Graduate Teaching Assistant, Introduction to Environmental Science (22 students)
2023	Teaching Assistant, Paleontology laboratory; 3rd year course (11 Students)
2023	Teaching Assistant, Earth System Processes; 1st year course (50 students)
2022	Teaching Assistant, Earth and Planetary Sciences; 1st year course (50 students)

Leadership skills

2025- present	Junior Executive, Student Government Association, QBioS, Georgia Tech
2025- Present	Instructor in Training (IIT) at Outdoor Recreation Georgia Tech (**ORGT**), **Caving**
2021-22	General Secretary, Games and Sports Student Affairs Council, IISER Kolkata - Two tenures, representing 1500 students.
2021	Fest Coordinator; Tenacity -- An all-IISER Virtual Games and Sports Fest

- Four-day online fest (during the pandemic) conducted among ten colleges with seven events.

2020 Department Representative; Batch of Geological Sciences 2023, IISER Kolkata.

Outreach Activities

2024-Present **Volunteer support to organizers**, **Fossil Friday**, Georgia Tech; help organize public outreach events that bring students and visitors hands-on with fossil specimens and paleontology research.

2022-2024 9th Indian National Exhibition cum- Fair and 25th National Science Exhibition Kolkata, India; **Member of outreach committee** of Earth Sciences (DES) IISER Kolkata

2022 **Founding Member**, Earth: The **Journal Club** of Earth Sciences, IISER Kolkata.

2021-2023 Organizer, **Green Bike Initiative**, IISER Kolkata; collected bikes abandoned around campus, auctioned some to students to fund repairs of others, and painted them green for students and visitors to use.r

2018- 2024 Member at Ek Pehal, an initiative to help educate kids in rural areas surrounding IISER Kolkata; organized multiple outreach events.